

Carbon-price inertia and cap-driven compliance in joint DDMRP strategic buffer positioning and supplier selection and allocation

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Abstract

This study develops an integrated optimization framework to test how carbon-price and emissions-cap instruments affect DDMRP strategic buffer positioning and multi-supplier selection and allocation. The modeled environmental boundary covers sourcing, inbound supplier transportation, internal transportation and warehousing emissions. Deterministic lead times and a guaranteed-service setting are considered for a single-product multi-echelon bill of materials. The nonlinear formulation is transformed into an explicit mixed-integer linear formulation by linearizing the products between buffer status, decoupled lead time and supplier-selection variables. A numerical evaluation examines scalability, the carbon-price (tax) sweep, emissions caps, the full-factorial combined carbon-price-and-cap design, BOM topology, multi-objective trade-offs, and near-optimal decision degeneracy, with the carbon price applied per tonne of CO₂ after correcting a gram-to-tonne monetization error in the earlier draft. The corrected results show price inertia: carbon prices of 15–100 EUR/tCO₂ do not alter emissions, buffer positions, supplier selections, or allocations, while an exploratory switching-threshold diagnostic finds the first price-only decision changes only at about 203–3227 EUR/tCO₂ across the representative instances. Binding emissions caps, in contrast, reduce emissions at a monotone but non-linear compliance cost, with the 70% cap requiring an 8–46% cost premium. Computationally, the native lexicographic baseline solves all scalability instances up to 150 components in at most 5.55 seconds, while 15 single-objective price, cap and combined PLM stress cases prove optimality in at most 1.58 seconds. A focused PLM/NLM validation attempts 21 small equivalence pairs; all 16 pairs that prove optimality in both formulations match within solver tolerance,

and five additional NLM incumbents match the PLM objective but do not prove optimality within the validation limit.

Keywords: DDMRP, Strategic buffer positioning, Supplier selection and allocation, Operational carbon emissions

1. Introduction

Supply-chain decision makers increasingly need to coordinate service, cost and environmental objectives. This creates a need for decision-support models that expose how sourcing and inventory choices interact with explicitly bounded operational emissions.

The article addresses this strategic decision problem by integrating DDMRP strategic buffer positioning with supplier selection and allocation. It models decision-relevant operational emissions from sourcing, inbound supplier transportation, internal transportation and warehousing, then evaluates whether carbon-price, emissions-cap and combined price-and-cap scenarios are strong enough to alter the resulting strategic operating point.

Demand Driven Material Requirements Planning emerged as a practitioner-driven evolution of MRP (Ptak and Smith, 2019). Recent optimization and review work automates buffer positioning, parameter tuning or operational DDMRP planning (Martin et al., 2023; Younespour et al., 2024; Habbadi et al., 2025; Muyulema-Allaica et al., 2025), while parallel research on carbon-regulated inventory and green supplier selection and order allocation embeds carbon prices, caps or cap-and-trade rules (Absi et al., 2013; Benjaafar et al., 2013; Choi, 2013; Dekker et al., 2012; Mirzaee et al., 2023; Hamdan et al., 2023; Shi and Zhu, 2025). Our earlier studies separately extended strategic positioning for hybrid MTO/MTS systems (Achergui et al., 2020, 2021a), carbon-aware positioning (Achergui et al., 2021b), and supplier selection (Achergui et al., 2022). Section 2.4 reviews these streams in detail. None of them, however, jointly optimizes multi-supplier selection and allocation with explicitly bounded operational carbon emissions under carbon-price, cap and combined price-and-cap scenarios within a guaranteed-service DDMRP strategic positioning model. The present article fills this gap by coupling decoupling-point placement with supplier allocation under explicit operational-emissions instruments. Its corrected numerical finding is deliberately instrument-specific: internationally informed per-tonne carbon prices

do not redirect the cost-optimal DDMRP–sourcing decisions, whereas binding emissions caps do and reveal the associated compliance-cost curve.

The remainder of the article is organized as follows. Section 2 defines the DDMRP strategic positioning problem, the multi-supplier allocation setting and the operational carbon-emissions boundary, and Section 2.4 reviews related literature. Section 3 presents the integrated nonlinear formulation and its mixed-integer linearization. Section 4 describes the solution approach. Section 5 reports the computational experiments and discusses the results. Section 6 concludes the study.

2. Problem definition

2.1. DDMRP method

Demand Driven Material Requirements Planning (DDMRP) is a multi-echelon planning and execution method that blends elements of MRP, Distribution Requirements Planning, lean and the theory of constraints to protect and promote the flow of materials and information Ptak and Smith (2019). Its central idea is to decouple the supply chain by placing strategically positioned inventory buffers that absorb demand and supply variability and compress the cumulative lead time seen by the customer. The method is organised around five sequential components: (i) strategic inventory positioning, which decides where buffers should be placed along the bill of materials (BOM); (ii) buffer profiles and levels, which size the buffers; (iii) dynamic adjustments, which adapt buffers to changing conditions; (iv) demand-driven planning, which generates replenishment orders; and (v) visible and collaborative execution.

Each buffer is described by three cumulative zones. The red zone provides safety against variability, the yellow zone covers demand over the decoupled lead time, and the green zone controls the replenishment frequency and order size. These zones are computed from the average daily usage (adu), the decoupled lead time of the node, a lead time factor ltf_i and a variability factor $varf_i$. Replenishment is then governed by the net flow equation, where the net flow position equals on-hand plus on-order inventory minus the qualified demand; an order is launched whenever this position drops into the yellow zone. This work addresses only the first component, strategic inventory positioning, because it conditions the decoupled lead times and therefore the inventory, cost and modeled emissions associated with each buffer.

2.2. Strategic inventory positioning

The placement of the decoupling points dictates the decoupled lead time a_i of every node of the BOM and, consequently, the average on-hand inventory aoi_i , the holding cost and the emissions of each potential buffer. Positioning too many buffers inflates the holding cost and the capital immobilization, whereas positioning too few lengthens the lead time exposed to the customer and jeopardizes the service commitment. Strategic inventory positioning therefore consists in selecting the subset of BOM nodes to buffer so that the promised customer service time SvT is honoured at the lowest possible cost.

We adopt the guaranteed service model (GSM) framework Graves and Willems (2000), in which every node quotes a guaranteed service time to its successors and is replenished within its decoupled lead time under deterministic assumptions. A buffered node decouples its predecessors and resets lead-time accumulation, while an unbuffered node propagates the lead time of its components. The qualitative DDMRP positioning factors indicate natural buffer candidates; this work casts strategic positioning as an optimization over binary buffer-placement variables x_i , coupled with supplier selection and allocation and the modeled operational emissions.

2.3. DDMRP strategic buffer positioning with multi-supplier allocation and operational carbon emissions

A deterministic multi-echelon supply-chain environment is considered. A final product $i = 1$ and its components $i \in N - \{0, 1\}$ are represented by a BOM, while a set of pre-selected eligible suppliers S replenishes raw materials and outsourced parts $i \in Dm$ under a promised service time SvT to customer node $i = 0$. The model addresses DDMRP strategic buffer positioning and its inventory implications; dynamic adjustment, demand-driven replenishment and execution are outside its scope. Manufacturing and delivery lead times are deterministic, production and storage capacities are assumed infinite, and supplier capacities are limited.

The models developed in Achergui et al. (2022), coupling buffer positioning with supplier selection, and in Achergui et al. (2021b), introducing emissions into the positioning problem, provide the basis for jointly accounting for supplier selection and allocation and the operational emissions of the modeled stages. The objective minimizes replenishment and storage costs along the BOM subject to service-time and supplier-capacity constraints. Emissions

sions are incorporated through a price in the objective, a cap constraint, or both simultaneously.

2.3.1. Carbon emissions modeling

The emissions function quantifies sources directly affected by buffer positioning and supplier selection and allocation. Within the adopted boundary, E is the sum of internal transportation emissions TE , warehousing emissions WE , and supplier-side emissions SE for externally delivered materials. This ordering–holding–transportation decomposition follows the structure used in green lot-sizing and SSOA models (Lamba and Singh, 2019; Toptal et al., 2014). Other lifecycle stages and Scope 3 categories are outside the model boundary (Matthews et al., 2008; Huang et al., 2009; U.S. EPA, 2022; Dekker et al., 2012). Let $X = \{x_i\}$, $\forall i \in N$ denote the buffer-positioning decisions.

$$E(X, Q) = \sum_{\forall i \in N} \sum_{\forall j \in S} E(x_i, q_{i,j}) = \sum_{\forall i \in N} [TE(x_i) + WE(x_i) + \sum_{\forall j \in S} SE(q_{i,j})] \quad (1)$$

Transportation emissions

Getting products from point A to point B is the main line of flow management. Internal between sites or external between echelons of the supply chain, transportation ensures this mission by a multi-modal service provision. Emissions related to external inventory movements by selected suppliers are included in supplier-side emissions SE detailed later in this section. On a strategic level, this study focuses on internal transportation between the sites that handle the BOM elements. The listing of every replenishment order would be complicated at this level of analysis. For simplification purposes, internal transportation emissions are considered for the average on hand inventory where each item i has a quantity of emissions β_i .

It is observed by the results of the inventory positioning models developed previously that decoupling points allow a reduction of replenishment orders frequency compared to non-decoupled points that had no inventory. This is related to the need for daily deliveries to compensate for the lack of stored parts. The carbon emissions emitted by replenishment activity are then reduced by a factor R for parts with buffers. This factor is related to the dimensions of the buffer, as it covers the maximum lead time of the product's components by which the replenishment is received. Based on previous simulations and the average lead time of the instances studied, for this work, we consider the factor to be equal to $R = 3$. The transportation emissions will

then be equal to $\beta^* = \beta/R$, for the parts where an inventory is positioned. The transportation emissions of an item i are expressed as follows:

$$TE(x_i) = \beta^* \cdot aoi_i \cdot x_i + \beta \cdot aoi_i \cdot (1 - x_i), \quad \forall i \in N \quad (2)$$

Storing emissions

These emissions occur in the event that a component is stored along the BOM. Setting up an inventory produces two sources of emissions: constant and variable emissions.

$$WE(x_i) = CWE(x_i) + VWE(x_i), \quad \forall i \in N \quad (3)$$

Constant emissions α^* are due to fixed operating resources in the storage facility, such as the energy consumed and the human resources needed. It could be product-specific in the case of condition-sensitive products, as they require special storage facilities. For this work, its amount is supposed to be the same for all items, as we consider the same storing conditions.

$$CWE(x_i) = \alpha^* \cdot x_i, \quad \forall i \in N \quad (4)$$

The variable emissions α_i are related to the amount of product stored in the facility and represent handling, deterioration and waste effects within the modeled warehousing boundary.

$$VWE(x_i) = \alpha_i \cdot aoi_i \cdot x_i, \quad \forall i \in N \quad (5)$$

Supplier-side emissions

Supplier-side emissions are represented by a fixed factor sup_Emis per allocated unit, including inbound transportation to the unloading dock. This factor allows cleaner suppliers to be selected despite a higher price when required by the active carbon scenario. With $Z = \{z_{i,j}\}$ the supplier-selection variables and $q_{i,j}$ the allocation variables, supplier-side emissions are formulated as follows:

$$SE(Q) = \sum_{\forall i \in Dm} \sum_{\forall j \in S} z_{i,j} \cdot q_{i,j} \cdot sup_Emis(j) \quad (6)$$

2.3.2. Carbon reduction strategies

The modeled emissions are integrated into the strategic buffer-positioning and supplier-selection-and-allocation problem through three policy-inspired

decision scenarios: a carbon price, an emissions cap, and a combined carbon-price-and-cap scenario.

The carbon tax strategy monetizes modeled emissions through the parameter *EmisTax*, expressed in constant 2023 EUR per tonne of CO₂ (2023 EUR/tCO₂), and adds this carbon cost to the objective function (Choi, 2013; He et al., 2015; Toptal et al., 2014). Explicit carbon prices are conventionally stated per tonne of CO₂ or CO₂ equivalent (World Bank, n.d.; OECD, 2025); there is no universal international tax rate because rates and coverage differ by jurisdiction. Since this model calculates emissions E in gCO₂, the corresponding objective term is $EmisTax \cdot E/10^6$. All purchasing, holding and carbon-price inputs must be expressed in constant 2023 EUR; Section 5 shows that this term is below the decision-switching threshold at the policy-informed values tested here.

The carbon cap limit strategy consists in simulating the regulation of emission target caps. Carbon emissions resulting from an activity cycle are limited, and it is up to the company to adapt its operational mode to respect them (Matthews et al., 2008; Lamba et al., 2019; Chowdhury, 2020). Within the modeled decision scope, this adaptation can occur by selecting lower-emission suppliers, reallocating quantities or repositioning buffers. To keep carbon emissions below this limit *ECap*, a constraint is added to the model that restricts the emission function. This could be considered as an emissions buffer for product’s BOM.

The combined carbon-price-and-cap scenario crosses the carbon price with either an explicit no-cap reference or an emissions cap imposed as a hard constraint. It is a policy-inspired decision scenario, not a representation of cap-and-trade, because permits, allowances and trading are not modeled (Mirzaee et al., 2023; Lamba et al., 2019; Rosyidi and Pratama, 2024). Cap-and-trade SSOA and lot-sizing models instead allow permit purchase and sale around a regulated ceiling (He et al., 2015; Rosyidi and Pratama, 2024). For capped levels, the operating point is shaped jointly by the cap stringency and emissions price.

2.4. Related literature

Four research streams frame the present study: DDMRP strategic positioning, carbon-regulated inventory planning, green supplier selection and order allocation (SSOA), and their integration under operational emission boundaries.

DDMRP strategic positioning and buffer optimization. Beyond practitioner manuals (Ptak and Smith, 2019), empirical assessments confirm DDMRP applicability in complex manufacturing (Miclo et al., 2019; Velasco Acosta et al., 2020), while a recent PRISMA-based review shows that strategic inventory positioning, buffer sizing and demand-driven planning remain distinct research themes with limited cross-integration (Muyulema-Allaica et al., 2025). Optimization studies cast buffer placement or parameter setting as mixed-integer programs, metaheuristics or simulation–optimization hybrids (Achergui et al., 2020, 2021a; Lahrichi et al., 2022; Damand et al., 2022; Martin et al., 2023; Habbadi et al., 2025; Xu et al., 2023; Younespour et al., 2024; Krajčovič et al., 2024). Favaretto et al. (2023) provide a theoretical validation of DDMRP reorder policies through time-varying (s, S) thresholds under service-level constraints. These contributions automate positioning or parameter tuning, and some extend carbon-aware positioning (Achergui et al., 2021b) or supplier selection (Achergui et al., 2022), but none jointly optimizes multi-supplier allocation with carbon-price, cap and combined price-and-cap scenarios within one guaranteed-service positioning model.

Carbon-regulated inventory and lot-sizing. Operations-research reviews identify inventory, transportation and sourcing as the main levers through which logistics decisions shape greenhouse-gas outcomes (Dekker et al., 2012). Lot-sizing and production-planning models incorporate carbon caps, taxes or cap-and-trade rules at item, period or network level (Absi et al., 2013; Benjaafar et al., 2013; Choi, 2013; Lamba et al., 2019; He et al., 2015; Toptal et al., 2014). Toptal et al. (2014) compare cap, tax and cap-and-trade policies in an EOQ setting with emission-reduction investment, while He et al. (2015) analyze lot-sizing trade-offs under tax and cap-and-trade in a *Journal of Cleaner Production* setting closely aligned with low-carbon manufacturing research. Because a tax enters these optimization models as an objective weight rather than as a hard feasibility restriction, its operational effect depends on whether the monetized emissions difference exceeds the procurement, holding and service-time trade-offs between feasible alternatives. Lamba and Singh (2019) and Chowdhury (2020) further decompose emissions from ordering, holding and transportation within multi-supplier lot-sizing MINLPs evaluated under alternative regulatory instruments. These studies clarify how regulatory instruments reallocate order quantities and replenishment timing, but they do not represent DDMRP decoupling points along a multi-echelon BOM nor link buffer status to replenishment-frequency

effects on transportation emissions.

Green supplier selection and order allocation. The multi-period, multi-product SSOA problem originates in lot-sizing models that jointly decide order quantities and supplier choice (Basnet and Leung, 2005). Recent SSOA models increasingly combine economic criteria with supplier-specific emission intensities, transport distances or disruption risks (Mirzaee et al., 2023; Hamdan et al., 2023; Shi and Zhu, 2025). Hamdan et al. (2023) formulate a bi-objective integer linear program for dynamic green SSOA with quantity discounts, while Shi and Zhu (2025) optimize procurement cost, quality losses and carbon emissions under disruption risk. Lamba et al. (2019) compare cap, tax and cap-and-trade regulations within a multi-period SSOA and lot-sizing MINLP, and Rosyidi and Pratama (2024) integrate transport-mode choice with cap-and-trade permit costs in a recent MILP. These models treat supplier choice and order split as primary decisions, but they omit strategic buffer placement on an assembly BOM and the GSM service-time logic that drives decoupled lead times in DDMRP positioning.

Integration gap. Prior work by the authors separates carbon-aware buffer positioning (Achergui et al., 2021b) from supplier-selection coupling (Achergui et al., 2022). Recent DDMRP optimization advances positioning, sizing or operational planning (Younespour et al., 2024; Krajčovič et al., 2024), and recent SSOA advances carbon-regulated procurement (Hamdan et al., 2023; Shi and Zhu, 2025; Rosyidi and Pratama, 2024), yet no published model unifies (i) binary decoupling-point decisions on a BOM, (ii) multi-supplier selection and quantity allocation, and (iii) policy-inspired carbon-price, cap and combined price-and-cap scenarios within one explicit mixed-integer framework bounded to sourcing, inbound supplier transportation, internal transportation and warehousing emissions. The combined scenario differs from cap-and-trade SSOA models (Mirzaee et al., 2023; Rosyidi and Pratama, 2024) by fixing permit markets outside the decision scope. The formulation developed in Section 3 addresses this gap.

The preceding subsections define the emissions boundary, the three carbon scenarios and their link to buffer placement and supplier decisions. Section 3 integrates these elements into a single strategic optimization model.

3. Problem formulation

The integrated strategic buffer-positioning and supplier-selection-and-allocation problem is first expressed as a nonlinear mixed-integer formulation. The

model accounts for decision-relevant operational carbon emissions within the sourcing, inbound transportation, internal transportation and warehousing boundary. An explicit linearization is then introduced to improve numerical tractability.

The developed model answers the following three questions at once:

- Select eligible suppliers and allocate the required quantities of raw materials and outsourced parts among them.
- Position inventories along the components of the BOM, in a combination where it does not cost much and does synchronize with supply capacities.
- Depending on the chosen strategy, monetize modeled emissions through a per-tonne carbon price or impose an emissions ceiling that can force lower-emission suppliers, quantity reallocations and buffer repositioning.

In the following, all the notation used for the mathematical formulation of the developed model is given in Table 1.

Table 1: Integrated buffer-positioning and supplier-selection-and-allocation notation.

Sets	Signification
N	Set of BOM nodes representing parts $i \in N$
Dm	Set of delivered materials and outsourced parts, $d \in Dm$
$Comp_i$	Components list of the i^{th} part, $k \in Comp_i$
S	Set of preselected suppliers for the BOM
SL_d	Set of accepted suppliers for the part d , $j \in SL_d \subseteq S$
Parameters	Signification
t_i	Processing time of the i^{th} part
ltf_i	Lead time factor of i^{th} part
$varf_i$	Variability factor of i^{th} part
adu	Average daily usage of the final product
$rqt f_i$	Required quantity of the i^{th} part to produce the final product
up_i	Unit price of the i^{th} part
pc_d	Processing costs of delivered material d
$aihc_i$	Average inventory holding cost rate of the i^{th} part

Continuation of Table 1		
$sup_Price_{d,j}$		Unit price of the j^{th} supplier for the part d
$sup_Deliv_{d,j}$		Delivery time of the j^{th} supplier for the part d
$sup_Cap_{d,j}$		Delivering capacity of the j^{th} supplier for the part d
$sup_Emis_{d,j}$		Emissions of the j^{th} supplier for the part d
α^*		Warehousing facilities emissions
α_i		Storing emissions of the part i
β_i		Transportation emissions of the part i
R		Replenishment emissions reduction factor
SvT		Promised customer service time
$ECap$		Emissions cap limit
$EmisTax$		Carbon price in constant 2023 EUR/tCO ₂
τ		Tax strategy indicator (1 if the carbon tax is active)
κ		Cap strategy indicator (1 if the carbon cap is active)
Decision Variables		Signification
x_i		Buffer status for the i^{th} part
a_i		Lead time of the i^{th} part
$z_{d,j}$		Supplier j status for the d^{th} part
$q_{d,j}$		Ordered quantity of the d^{th} part from the supplier j
$v_{d,j}$		Linearization variable equal to $y_d z_{d,j}$
Decision Expres- sions		Signification
SvT_0		Actual service time
aoi_i		Average on-hand inventory of the i^{th} part
yz_i		Buffer yellow zone of the i^{th} part
gz_i		Buffer green zone of the i^{th} part
rz_i		Buffer red zone of the i^{th} part
StC_i		Storing cost of the i^{th} part
$RawMC_{d,j}$		Raw material cost of the d^{th} part from the supplier j
TC		Total of Storing costs
E		Total carbon emissions of the BOM
TE		Transportation emissions
WE		Warehousing emissions
SE		Supplier-side emissions

The model outputs the buffer positions, supplier selections and allocations, and the decision-relevant operational emissions E . Its final expression

is formulated using Equation 2, Equation 3 and Equation 6 as follows:

$$E(X, Q) = \sum_{\forall i \in N} [[\alpha^* + (\alpha_i + \beta_i^* - \beta_i) \cdot aoi_i] \cdot x_i + \beta_i \cdot aoi_i + \sum_{\forall j \in S} q_{i,j} \cdot sup_Emis_j] \quad (7)$$

For the **Carbon emissions Tax** strategy, the emissions compensation cost is added to the objective function. It would be minimized in parallel to storing costs and purchasing costs of raw materials. The model in this case is formulated as follows:

$$\begin{aligned} \min TC &= RawMCost + \sum_{\forall i \in N} StC_i + EmisTax \cdot \frac{E(X, Q)}{10^6} \quad (8) \\ S.T. \\ SvT_0 &\leq SvT \\ a_i &= t_i + \max_{k \in comp_i} [(1 - x_k) \cdot a_k], \quad \forall i \in N \setminus Dm \\ a_d &= t_d + \max_{j \in SL_d} z_{d,j} \cdot sup_Deliv_{d,j}, \quad \forall d \in Dm \end{aligned}$$

The Carbon cap limit strategy holds the total emissions at the limit of $ECap$ which is added as a second hard constraint to the model in addition to the service time. This limit is considered a reasonable cap to allow some maneuverability in managing the different operations. The model for this strategy is expressed as follows:

$$\begin{aligned} \min TC &= RawMCost + \sum_{\forall i \in N} StC_i \quad (9) \\ S.T. \\ SvT_0 &\leq SvT \\ E(X, Q) &\leq ECap \\ a_i &= t_i + \max_{k \in comp_i} [(1 - x_k) \cdot a_k], \quad \forall i \in N \setminus Dm \\ a_d &= t_d + \max_{j \in SL_d} z_{d,j} \cdot sup_Deliv_{d,j}, \quad \forall d \in Dm \end{aligned}$$

4. Solution approach

This section presents the solution approach for the integrated DDMRP strategic buffer-positioning and supplier-selection-and-allocation problem. The

original formulation contains products between binary buffer or supplier decisions and integer decoupled lead times. Following Achergui et al. (2021a), these products are replaced by explicit linearization variables, yielding a mixed-integer linear formulation solved with CPLEX.

The big- M method replaces $a_i x_i$ with y_i and $y_i z_{i,j}$ with $v_{i,j}$. The bound M is derived from the sum of processing times and the maximum eligible supplier lead time. Supplier-selection variables are linked to strictly positive allocations and supplier eligibility. Two binary scenario indicators control which carbon mechanism is active: τ activates the carbon-price term in the objective function, and κ activates the emissions-cap constraint. The three scenarios are recovered as special cases of a unified model:

$$(\tau, \kappa) = \begin{cases} (1, 0), & \text{Tax strategy} \\ (0, 1), & \text{Cap limit strategy} \\ (1, 1), & \text{Hybrid strategy} \end{cases} \quad (10)$$

Let $H_i = adu \cdot aihc_i \cdot (1.5 + varf_i) \cdot ltf_i \cdot up_i \cdot rqt f_i$. Using Equation 7, the

linearized model is written compactly as follows:

$$\min TC = RawMCost(Q) + \sum_{\forall i \in N} H_i \left[y_i + \sum_{\forall j \in S} su_{i,j} sup_price_j v_{i,j} \right] + \tau EmissionTax \frac{E(X, A, Y, Q)}{10^6} \quad (11)$$

S.t.

$$SvT_0 \leq SvT \quad (12)$$

$$\kappa \cdot E(X, Q) \leq ECap \quad (13)$$

$$0 \leq a_i \leq M, \quad \forall i \in N \quad (14)$$

$$0 \leq y_i \leq a_i, \quad \forall i \in N \quad (15)$$

$$y_i \leq M \cdot x_i, \quad \forall i \in N \quad (16)$$

$$y_i \geq a_i - M \cdot (1 - x_i), \quad \forall i \in N \quad (17)$$

$$0 \leq v_{i,j} \leq y_i, \quad \forall i \in N, \forall j \in S \quad (18)$$

$$v_{i,j} \leq M z_{i,j}, \quad \forall i \in N, \forall j \in S \quad (19)$$

$$v_{i,j} \geq y_i - M(1 - z_{i,j}), \quad \forall i \in N, \forall j \in S \quad (20)$$

$$z_{d,j} \leq q_{d,j} \leq z_{d,j} su_{d,j} sup_cap_{d,j}, \quad \forall d \in Dm, \forall j \in S \quad (21)$$

$$z_{d,j} \leq su_{d,j}, \quad \forall d \in Dm, \forall j \in S \quad (22)$$

$$\sum_{\forall j \in S} q_{d,j} = adu \cdot rqtfd_d, \quad \forall d \in Dm \quad (23)$$

$$a_i \geq t_i + (1 - x_k) \cdot a_k, \quad \forall i \in N \setminus Dm, \quad \forall k \in comp_i \quad (24)$$

$$a_d \geq t_d + z_{d,j} \cdot sup_deliv_{d,j}, \quad \forall d \in Dm \quad (25)$$

We maintain throughout this work the same resolution strategy as used in our previous models Achergui et al. (2021a, 2022). The proposed model is solved in parallel with its non-linear version. This solution approach allows a benchmark of the results obtained by the pseudo-linearization. The behavior of the model around different types of BOMs is also a key data for future perspectives of this research work.

5. Experimentation

The numerical evaluation covers computational scalability, carbon-policy scenarios, service-time sensitivity, multi-level and parallel BOM topologies, multi-objective Pareto analysis, near-optimal decision-degeneracy diagnostics, and a comparison of the linearized formulation with its nonlinear coun-

terpart. Following a unit audit, the carbon price is applied as *EmisTax* in currency per tonne of CO₂, with model emissions expressed in grams and converted to tonnes by dividing by 10⁶ before the price enters the cost objective. The earlier per-gram interpretation, which inflated every tested carbon price by a factor of 10⁶, has been corrected, and all experiment families—scalability, the carbon-price (tax) sweep, the cap sweep, the full-factorial combined carbon-price-and-cap design, service-time sensitivity, topology sensitivity, Pareto analysis, the near-optimal decision-degeneracy diagnostics and the PLM/NLM validation—are reported on this corrected basis. CPLEX 20.1.0 solves the linearized formulation and CP Optimizer solves the nonlinear formulation on a machine with an Intel i7-12700K 3.61 GHz and 32GB of RAM running Windows 10, using four solver threads. A 300-second limit is imposed on main campaign runs; the focused PLM/NLM equivalence validation reported below uses a 60-second validation limit. Solver termination status and optimality gaps are retained so that proven-optimal solutions are distinguished from time-limited incumbents. Time-limited NLM incumbents are reported when available but are not treated as bit-reproducible proof of optimality. All runs remain reported, but behavioral comparisons admit only proven-optimal solutions and time-limited feasible incumbents with a final optimality gap no greater than 1%; larger-gap incumbents are excluded from comparative conclusions. For representative proven-optimal scenarios, additional near-optimal probes maximize buffer-position, supplier-selection and allocation deviations while constraining the original objective to remain within 1% of the optimum; these probes diagnose whether a small objective gap hides materially different decisions without multiplying the full policy sweep. The resulting Jaccard similarities and allocation deviations are reported as continuous diagnostics rather than stable/unstable classes, because no industrial tolerance threshold is imposed. They are not interpreted as stochastic robustness tests, which would require demand variability, supplier disruption, or execution-dynamics experiments. Throughout this section, emissions are reported in tonnes of CO₂ (1 t = 10⁶ gCO₂).

5.1. Benchmark instances and simulation scenarios

The supply chain network environment considered previously is kept to ensure the accuracy of the generated instances. The benchmark set spans BOM sizes from $N = 2$ to $N = 150$ components and includes both tree-structured assemblies and dedicated multi-level and parallel-branch topologies. In this section, simulation scenarios denote controlled deterministic

optimization scenarios, not Monte Carlo, stochastic demand, disruption, or dynamic execution simulations. They test the three carbon-policy settings for the BOMs considered. The corrected carbon-price comparison uses a no-price baseline and $EmisTax \in \{15, 50, 75, 100\}$ constant 2023 EUR/tCO₂. These internationally informed levels span a low observed carbon-tax signal and progressively stronger policy scenarios, but they are study-defined comparison values rather than statutory rates or a direct reproduction of a single international benchmark (OECD, 2025; High-Level Commission on Carbon Prices, 2017). The emissions baseline used to set cap percentages is selected with CPLEX’s native lexicographic objective, $staticLex(TotalCostCS, Emis)$: economic cost is minimized first with no carbon price and no cap, and supplier-side emissions are then minimized as the internal tie-break objective. This baseline objective is used only to define reproducible cap ceilings; policy runs minimize the active scenario objective $TotalCostTS + \sum_i a_i$ with the relevant price and/or cap terms. At $EmisTax = 0$, the 100% cap reproduces the baseline emissions ceiling, while the no-cap and 100%-cap levels remain distinct experimental design points. The emission cap $ECap$ is tightened progressively from 100% down to 70% of this reproducible baseline. The same suppliers details seen previously are considered with the same adaptation for their capacities to accommodate the higher demand of complex BOMs. A supplier-side emissions quantity per product is added to these details as discussed in the previous section. Their details are updated and presented in table 2.

We assume that internal trips are made on the factory premises using the same vehicle. For an average travel distance, we fix the amount of transportation emissions generated in a whole activity cycle to $\beta = 1500gCO_2/unit$. Supplier-side emissions include the transportation emissions of the selected suppliers until the unloading dock. The supplier table does not impose a universal emissions–price functional relationship; however, in this calibrated set several low-price suppliers are also relatively low-emission options, which matters for interpreting the carbon-price inertia reported below. For simplification purposes, the pre-selected suppliers S are the same for all the components of the BOMs, and all supplier related detail for a non-delivery material is null.

5.2. Results and analysis

The corrected analysis reports every experiment family on the per-tonne carbon-price basis. The key outputs are the actual service time SvT_0 , total

Table 2: Details of the Suppliers list

s	1	2	3	4	5	6	7	8	9	10
$sup_deliv_{d,j}$	2	2	1	1	5	6	7	5	9	1
$sup_price_{d,j}(\%)$	100	90	110	100	60	50	50	30	20	80
$sup_cap_{d,j}$ simp.	100	60	80	80	50	60	80	25	250	30
$sup_cap_{d,j}$ comp.	700	800	680	480	550	650	1100	700	1700	5800
$sup_Emis_{d,j}$	80	75	90	100	40	45	40	30	15	40

s	11	12	13	14	15	16	17	18	19	20
$sup_deliv_{d,j}$	10	2	1	3	5	6	7	4	9	10
$sup_price_{d,j}(\%)$	20	60	80	35	30	40	15	30	20	10
$sup_cap_{d,j}$ simp.	300	60	50	35	50	20	100	80	150	300
$sup_cap_{d,j}$ comp.	1800	400	300	250	350	550	1000	450	950	1400
$sup_Emis_{d,j}$	15	50	80	60	30	5	30	40	30	80

cost TC , decision-relevant operational carbon emissions E , number of positioned buffers, days of inventory outstanding (DIO, computed as the sum of decoupled lead times), computational time RT , solver status and optimality gap. All runs are reported; behavioral comparisons draw only on comparison-admissible solutions, that is proven-optimal solutions and time-limited feasible incumbents whose final optimality gap does not exceed 1%.

5.2.1. Computational scalability

Table 3 reports the behavior of the pseudo-linear model on a baseline scenario (no carbon policy) for BOM sizes ranging from $N = 2$ to $N = 150$ components. All instances are solved to optimality, and the solving time stays on the order of a second across the whole range: most instances solve in well under one second, the largest ($N = 150$) in under one second, and the slowest cases ($N = 35$ and $N = 123$) in about 5.5 seconds. This confirms that the pseudo-linearization keeps the integrated positioning and supplier-selection problem tractable at an industrial scale, well within the time windows required for strategic and tactical planning. As illustrated in Figure 1, the runtime does not grow monotonically with N : it is driven by the combinatorial structure of each BOM (its depth, branching and the resulting supplier-selection sub-problem) rather than by the node count alone.

The number of positioned buffers and the resulting DIO grow with BOM complexity (Figure 2), from a single buffer for the smallest chains up to 57 buffers for the largest assembly. Here again the relation is not strictly monotonic in N : instances with a flatter or more parallel structure concentrate the

Table 3: Scalability of the pseudo-linear model across BOM sizes (baseline, zero carbon tax).

N	Buffers	DIO	Emissions (t CO ₂)	Runtime (s)	Status
2	1	4	0.11	0.020	OPTIMAL
3	2	8	0.24	0.030	OPTIMAL
4	3	12	0.46	0.160	OPTIMAL
5	3	31	2.93	0.160	OPTIMAL
6	4	29	2.68	0.700	OPTIMAL
7	4	35	3.85	0.290	OPTIMAL
8	3	49	5.59	0.110	OPTIMAL
10	4	74	9.95	0.330	OPTIMAL
13	7	135	29.5	0.160	OPTIMAL
15	8	158	33.1	0.190	OPTIMAL
20	6	249	25.4	1.550	OPTIMAL
25	6	320	31.2	1.690	OPTIMAL
26	1	431	41.6	0.190	OPTIMAL
30	1	479	49.6	0.390	OPTIMAL
35	9	480	48.3	5.550	OPTIMAL
40	5	532	101.7	0.320	OPTIMAL
50	5	707	194.6	0.480	OPTIMAL
60	5	1139	1 788	0.860	OPTIMAL
80	5	1454	2 591	0.390	OPTIMAL
90	14	1798	9 093	0.300	OPTIMAL
100	18	2251	14 450	0.220	OPTIMAL
123	7	1906	647.0	5.310	OPTIMAL
150	57	3126	25 080	0.990	OPTIMAL

decoupling on very few nodes, whereas deeper assemblies require more buffers to protect the promised service time. The baseline emissions reported in Table 3 and visualized in Figure 3 follow the same structural dependence; they are governed by the quantities flowing through the BOM and by the selected suppliers, so they scale with complexity while varying between instances of comparable size.

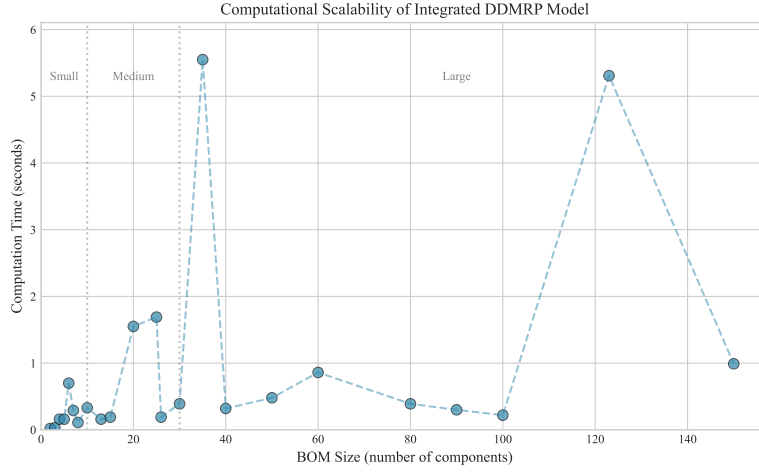


Figure 1: Computational time of the pseudo-linear model across BOM sizes.

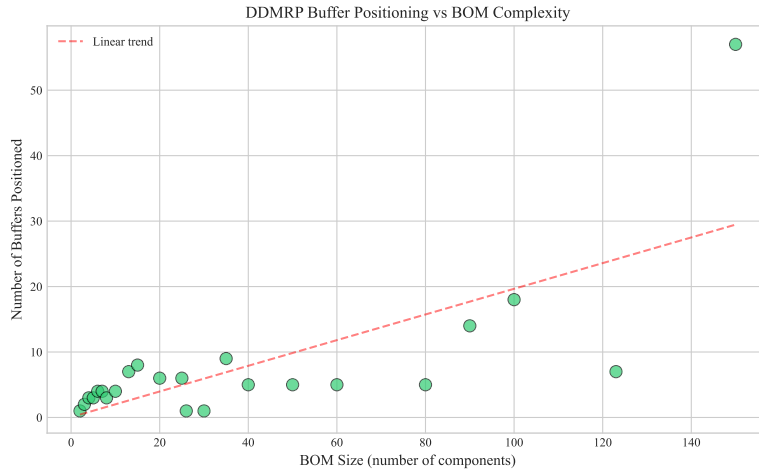


Figure 2: Number of positioned DDMRP buffers across BOM sizes.

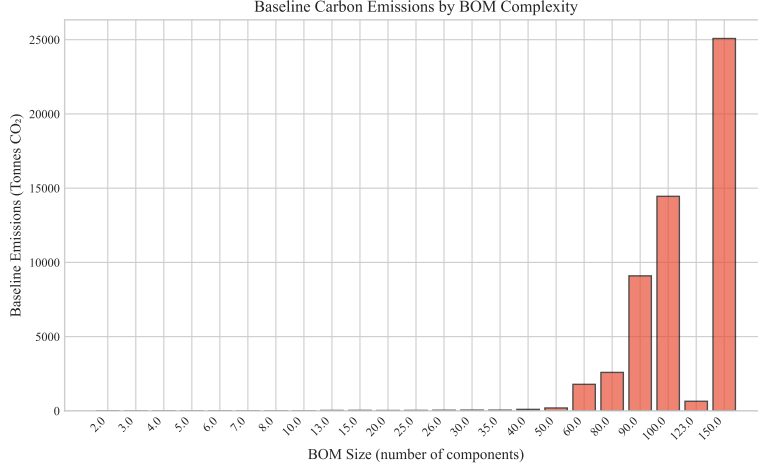


Figure 3: Baseline carbon emissions (no carbon policy) by BOM complexity.

5.2.2. Carbon tax strategy

Under the carbon tax strategy, monetized emissions are added to the cost objective, so the model trades off purchasing and holding costs against the emission penalty $EmisTax \cdot E$, with E expressed in tonnes. Table 4 reports the total emissions of the six representative instances across the corrected carbon-price comparison levels $EmisTax \in \{0, 15, 50, 75, 100\}$ constant 2023 EUR/tCO₂. Over this range the emissions—and the underlying buffer positions and supplier allocations—are invariant: for every tested instance the highest price yields exactly the baseline emissions, a 0% reduction. At these internationally informed price levels the monetized carbon term stays small relative to the procurement- and inventory-cost differences between alternative sourcing and buffering decisions, so it does not displace the cost-optimal solution. This is the substantive correction relative to the earlier per-gram draft, in which the price was overstated by a factor of 10^6 and therefore appeared to reshape the decisions: once the price is applied per tonne, a hard emissions cap, rather than the price, is what drives the emission response over the tested range.

To test whether this inertia is a genuine decision-threshold effect rather than a coding artifact, an exploratory switching-threshold diagnostic increases $EmisTax$ beyond the policy-informed range and locates the first price interval at which the price-only solution differs from the no-price operating point. Table 5 shows that all representative instances eventually switch sup-

Table 4: Carbon tax sweep: total emissions (t CO₂) per instance as the carbon price *EmisTax* (EUR/tCO₂) increases; column headings report *EmisTax* values.

Instance	0	15	50	75	100	Red. %
bom_5	2.93	2.93	2.93	2.93	2.93	0.0
bom_13	29.5	29.5	29.5	29.5	29.5	0.0
bom_26	41.6	41.6	41.6	41.6	41.6	0.0
bom_50	194.6	194.6	194.6	194.6	194.6	0.0
bom_ml4_30	13.6	13.6	13.6	13.6	13.6	0.0
bom_par4	5.94	5.94	5.94	5.94	5.94	0.0

pliers and allocations; in every switching case, the changed operating-point components are the supplier set, allocation quantities, tax-free cost and emissions, so this repeated diagnostic field is stated here rather than duplicated as a table column. For the regular BOM instances, the first switching intervals are about 607–609 EUR/tCO₂ for **bom_5**, 926–928 EUR/tCO₂ for **bom_13**, 2383–2391 EUR/tCO₂ for **bom_26**, and 203 EUR/tCO₂ for **bom_50**; the structural **bom_ml4_30** and **bom_par4** cases switch at about 3219–3227 and 656–658 EUR/tCO₂, respectively. These values should not be interpreted as realistic statutory taxes. They quantify the model’s internal price responsiveness: at observed comparison levels the carbon term is too small to overcome tax-free cost differences, while higher stress-test prices eventually trigger lower-emission supplier portfolios. The supplier table helps explain the inertia: several low-price suppliers are also relatively low-emission options, so economic-cost minimization already selects emissions-competitive suppliers in parts of the feasible set.

Table 5: Exploratory carbon-price switching-threshold diagnostic. The interval reports the first *EmisTax* range, in EUR/tCO₂, where the price-only operating point differs from the no-price solution; values above observed policy levels are stress-test diagnostics, not proposed statutory taxes.

Instance	Switching interval	Δ cost	Emission reduction (t CO ₂)
bom_5	607.4–609.4	175	0.29
bom_13	925.8–927.7	3 108	3.35
bom_26	2 382.8–2 390.6	1 960	0.82
bom_50	202.5–203.1	220	1.08
bom_ml4_30	3 218.8–3 226.6	1 400	0.43
bom_par4	656.2–658.2	96	0.14

5.2.3. Carbon cap strategy

Under the cap strategy, a hard constraint forces the total emissions below a ceiling $ECap$, expressed as a percentage of the baseline emissions. Table 6 reports the total cost as the cap is tightened from 100% down to 70% of the baseline, and Figure 4 shows the resulting compliance-cost curves. As expected, the cost increases monotonically with the stringency of the cap, since the model is forced to adopt cleaner but more expensive suppliers and to reposition buffers to meet the emission target. The compliance cost is markedly non-linear: it remains modest for caps down to about 85% of the baseline and rises sharply for the tightest caps. Over the tested instances, the cost premium at the 70% cap ranges from roughly 8% to 46%, so the cap experiment provides the main actionable policy signal: binding ceilings can redirect strategic sourcing and buffering, but overly tight targets can create disproportionate compliance costs.

Table 6: Carbon cap sweep: total cost per instance as the emission cap tightens from 100% to 70% of the baseline emissions.

Instance	100%	95%	90%	85%	80%	75%	70%
bom_5	48 640	48 815	51 555	51 555	51 555	60 260	60 260
bom_13	116 450	118 410	119 560	127 100	127 890	151 640	170 520
bom_26	867 120	874 160	888 160	909 180	931 800	957 140	1 008 200
bom_50	1 381 500	1 402 900	1 436 300	1 497 800	1 566 100	1 601 800	1 637 000
bom_ml4_30	378 440	382 640	408 850	424 400	450 800	455 600	469 360
bom_par4	169 920	170 150	175 180	175 190	175 180	184 120	184 120

5.2.4. Hybrid strategy

The combined-scenario experiment crosses $EmisTax$ with either no cap or a hard emissions cap. The corrected experiment uses a full factorial design, testing every carbon-price comparison level against an explicit no-cap level and hard ceilings from 100% to 70% of baseline emissions. The 100% level remains a hard baseline-emissions ceiling and is distinct from no cap. Infeasible combinations are retained and reported rather than omitted or relaxed, because they identify the achievable emissions frontier; in the present campaign all 160 tested combinations are feasible and solved to proven optimality. Consistent with the carbon-price result, the achieved emissions are invariant to $EmisTax$ at every cap level—the cap alone governs them, decreasing monotonically as the ceiling tightens from the baseline down to 70%—while the price leaves the cost-optimal buffer positions and supplier

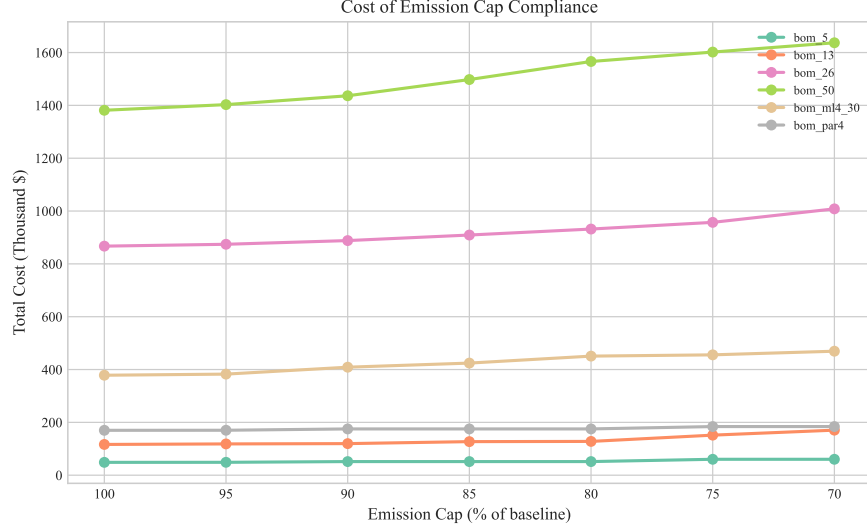


Figure 4: Compliance cost as the emission cap is tightened from 100% to 70% of the baseline.

allocations unchanged. This design distinguishes unconstrained price-only behavior, compliance at the baseline ceiling, and increasingly stringent cap effects; the complete per-combination results are tabulated in Appendix Appendix A (Table A.9).

5.2.5. Strategy comparison and cost–emissions trade-off

Across the corrected strategies the comparison is now unambiguous over the tested range: the carbon-price scenario does not alter the cost-optimal buffer positions or supplier allocations at the internationally informed per-tonne prices, the emissions-cap scenario reduces emissions at a monotonically increasing compliance cost, and the combined carbon-price-and-cap scenario inherits its emission behavior entirely from the cap. The separate multi-objective study makes the underlying cost–emissions trade-off explicit. Using the ε -constraint method on total cost and total emissions, the Pareto fronts of Figure 5 are obtained for three representative instances. A similar trade-off appears between cost and days of inventory outstanding in Figure 6. These fronts quantify feasible operating-point choices independently of the corrected carbon-price calibration.

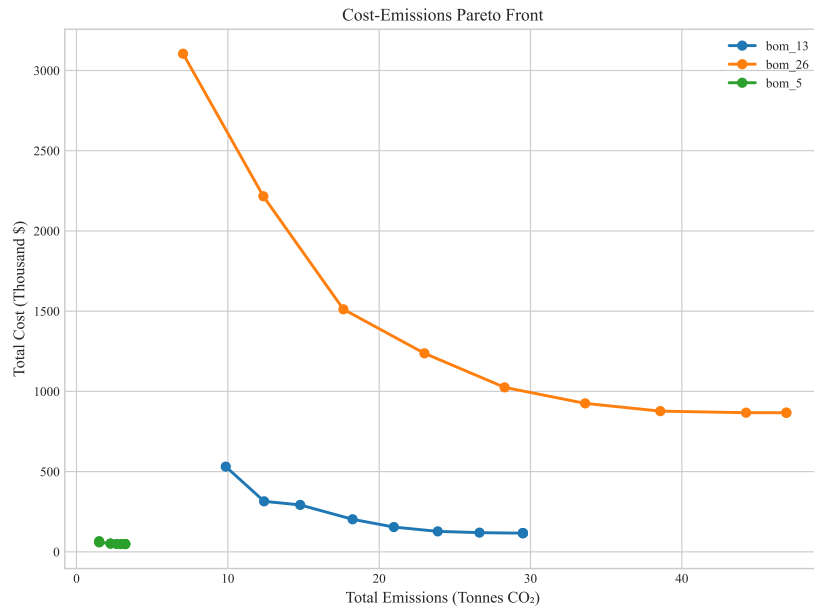


Figure 5: Cost-emissions Pareto fronts obtained by the ε -constraint method.

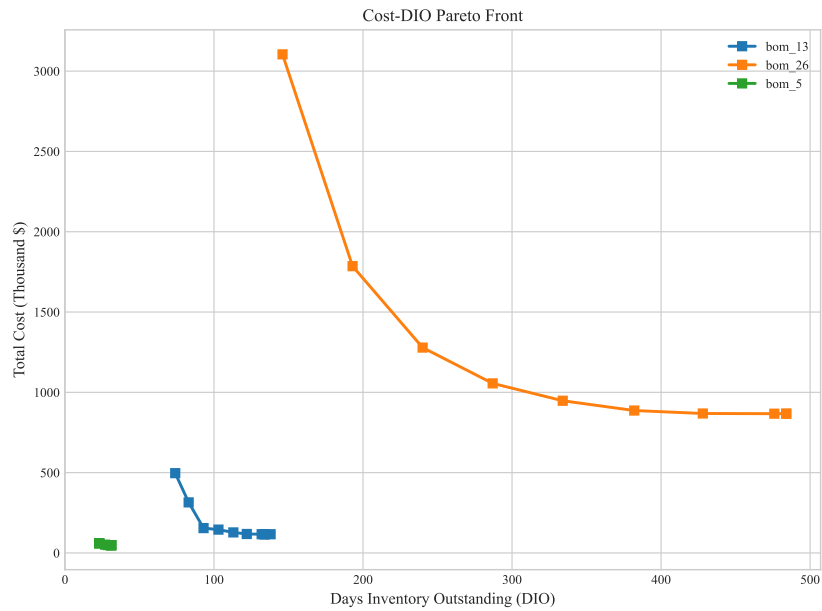


Figure 6: Cost-DIO Pareto fronts obtained by the ε -constraint method.

5.2.6. Near-optimal decision-degeneracy diagnostic

For representative proven-optimal policy anchors, extremal alternative solutions are computed that maximize the deviation in buffer positions, supplier selections and allocation quantities while constraining the original objective to remain within 1% of the optimum. Table 7 reports the resulting continuous similarity and deviation metrics—the minimum buffer-position and supplier-selection Jaccard similarities and the maximum normalized allocation L_1 deviation—for each anchor; no binary stable/unstable label is assigned and no industrial tolerance threshold is imposed. Across the anchors the buffer positions remain comparatively similar (buffer Jaccard frequently above 0.5 and reaching 1), whereas the supplier selections and allocation quantities vary substantially (supplier Jaccard as low as about 0.18, and normalized allocation deviation exceeding 1 for the smallest instances). The diagnostic therefore indicates that the cost-optimal objective value is attainable through materially different sourcing and allocation decisions while the decoupling structure stays comparatively concentrated. These are continuous diagnostics of decision degeneracy run on representative anchors only; they are not interpreted as stochastic robustness tests, which would require demand variability, supplier-disruption or execution-dynamics experiments.

5.2.7. Service-time and topology sensitivity

The service-time experiment varies the promised service time $SvT_0 \in \{0, 1, 3\}$ under both the carbon-price and the cap strategy (Figure 7). Relaxing the promised service time leaves the achieved emissions essentially unchanged but lowers both the total cost and the number of positioned buffers, because a longer admissible decoupled lead time lets the model protect the promised time with fewer buffers; consistent with the carbon-price result, the price level does not alter these decisions. The topology analysis of Figure 8, which uses the zero-price baseline, confirms that the model handles different BOM structures: across the multi-level and parallel-branch families, the baseline emissions and buffer placement adapt to the depth and branching of the BOM rather than to its size alone, and all topologies remain solved to optimality within the time limit.

5.2.8. Pseudo-linear versus non-linear model

Finally, the corrected-unit linearized formulation (PLM) is benchmarked against the nonlinear formulation (NLM) solved with CP Optimizer under a common 60-second validation limit. The focused equivalence benchmark

Table 7: Near-optimal decision-degeneracy diagnostic probes. Each row summarizes extremal alternatives whose original objective remains within 1% of the proven optimum. Source abbreviations are tax, cap and hybrid; no binary stability class is assigned.

Instance	Source	Tax	Cap	Buf. J	Sup. J	Alloc. L1
bom_5	cap	0.00	100%	0.75	0.22	1.52
bom_5	cap	0.00	85%	0.80	0.67	0.92
bom_5	cap	0.00	70%	0.83	0.60	0.89
bom_5	hybrid	15.00	none	0.75	0.22	1.53
bom_5	hybrid	50.00	85%	0.80	0.67	0.92
bom_5	hybrid	100.00	70%	0.83	0.60	0.89
bom_5	tax	0.00	none	0.75	0.22	1.52
bom_5	tax	50.00	none	0.75	0.22	1.54
bom_5	tax	100.00	none	0.75	0.22	1.57
bom_13	cap	0.00	100%	0.78	0.31	0.62
bom_13	cap	0.00	85%	0.60	0.34	0.64
bom_13	cap	0.00	70%	0.67	0.59	0.40
bom_13	hybrid	15.00	none	0.78	0.25	0.88
bom_13	hybrid	50.00	85%	0.60	0.34	0.65
bom_13	hybrid	100.00	70%	0.67	0.57	0.65
bom_13	tax	0.00	none	0.75	0.25	0.89
bom_13	tax	50.00	none	0.75	0.26	0.85
bom_13	tax	100.00	none	0.78	0.26	0.81
bom_26	cap	0.00	100%	0.50	0.21	0.32
bom_26	cap	0.00	85%	0.50	0.71	0.28
bom_26	hybrid	15.00	none	0.50	0.21	0.32
bom_26	hybrid	100.00	70%	0.33	0.50	0.33
bom_26	tax	0.00	none	0.50	0.21	0.32
bom_26	tax	50.00	none	0.50	0.21	0.32
bom_26	tax	100.00	none	0.50	0.21	0.33
bom_50	cap	0.00	100%	0.83	0.19	0.43
bom_50	cap	0.00	85%	0.83	0.55	0.32
bom_50	cap	0.00	70%	1.00	0.18	0.05
bom_50	hybrid	15.00	none	0.83	0.18	0.43
bom_50	hybrid	50.00	85%	1.00	0.18	0.09
bom_50	hybrid	100.00	70%	1.00	0.18	0.68
bom_50	tax	0.00	none	0.83	0.18	0.43
bom_50	tax	50.00	none	0.83	0.18	0.43
bom_50	tax	100.00	none	0.83	0.18	0.44

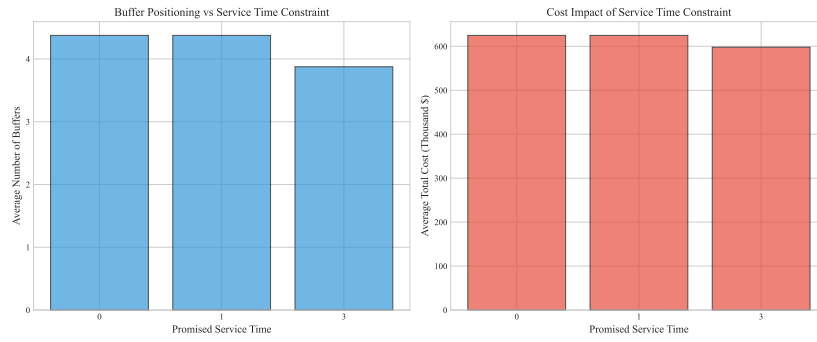


Figure 7: Service-time sensitivity of cost and positioned buffers under the carbon-price and cap strategies.

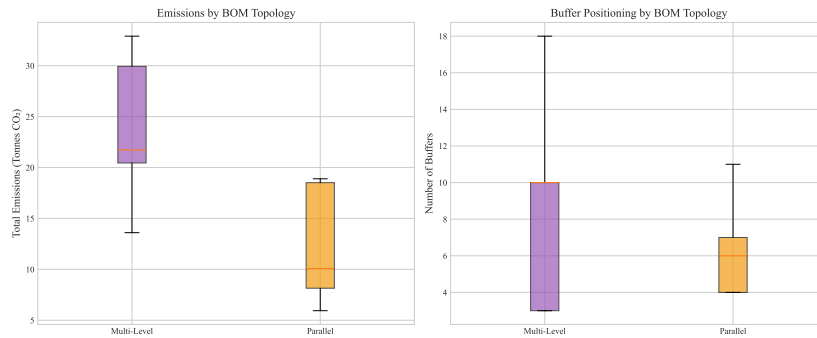


Figure 8: Baseline emissions across multi-level and parallel BOM topologies.

attempts 21 small PLM/NLM pairs: seven instance–scenario pairs for each of the price, cap and combined settings. Sixteen pairs terminate optimally in both formulations and all 16 match within an absolute objective tolerance of 1. Five additional NLM runs return time-limited feasible incumbents equal to the PLM objective but cannot prove optimality within the validation limit. Separately, all 15 larger single-objective PLM stress cases up to 150 components terminate optimally in at most 1.58 seconds. This stress-case runtime is not the same quantity as the native lexicographic scalability baseline in Table 3, whose slowest multi-priority baseline solve is 5.55 seconds. These results support empirical validation of the linearized formulation over the tested domain, rather than a claim of universal mathematical equivalence.

Table 8: Corrected-unit formulation validation. The equivalence benchmark attempts seven small PLM/NLM pairs per scenario under a 60 s NLM validation limit; matching counts include only pairs that prove optimality in both formulations. Time-limited equal incumbents match the PLM objective but do not prove NLM optimality.

Scenario	Attempted	Joint opt.	Matches	Equal inc.	Stress / max RT
Carbon price	7	4	4	3	5/5 / 1.58 s
Emissions cap	7	6	6	1	5/5 / 1.55 s
Combined price and cap	7	6	6	1	5/5 / 1.31 s
Total	21	16	16	5	15/15 / 1.58 s

6. Conclusion

This article proposes an integrated sustainable DDMRP decision framework that jointly optimizes strategic buffer positioning, supplier selection and quantity allocation under explicitly modeled operational emissions. Relative to recent DDMRP optimization studies that automate buffer placement or integrate strategic and operational planning (Martin et al., 2023; Younespour et al., 2024; Habbadi et al., 2025; Krajčovič et al., 2024), to carbon-regulated lot-sizing and inventory models (He et al., 2015; Toptal et al., 2014; Chowdhury, 2020), and to green supplier-selection models that optimize order allocation under carbon regulation (Mirzaee et al., 2023; Hamdan et al., 2023; Shi and Zhu, 2025; Lamba et al., 2019; Rosyidi and Pratama, 2024), the contribution lies in coupling DDMRP decoupling-point placement with multi-supplier allocation and an explicitly bounded operational-emissions function. The model minimizes cost while satisfying the promised deterministic service-time threshold and the active carbon-scenario conditions. Its environmental

conclusions are limited to the modeled boundary of sourcing, inbound supplier transportation, internal transportation and warehousing.

The corrected numerical evidence is instrument-specific. Carbon prices of 15, 50, 75 and 100 EUR/tCO₂ are correctly monetized but do not redirect the cost-optimal buffer positions, supplier selections or allocations in any representative instance. The exploratory threshold diagnostic shows that price-only decisions begin to change only at higher stress-test prices, from about 203 EUR/tCO₂ to above 3200 EUR/tCO₂ depending on the BOM and supplier-capacity structure. This price inertia is a decision-relevant finding rather than a modeling failure: at the tested policy-informed levels, the carbon-price term is below the endogenous switching threshold of the modeled cost-emissions trade-offs.

Binding caps, in contrast, are the operative emission-reduction mechanism over the tested range. They force lower-emission sourcing and allocation choices at a monotone but non-linear compliance cost; the cost premium remains moderate down to about the 85% cap and rises sharply for the tightest levels, reaching roughly 8–46% at the 70% cap. This provides a practical policy and managerial message: current-level price signals may be insufficient to redirect strategic DDMRP-sourcing decisions in this calibrated setting, while binding ceilings can do so but require careful calibration to avoid disproportionate compliance costs.

The near-optimal decision-degeneracy diagnostic qualifies the prescriptive interpretation. Buffer positions are comparatively concentrated across alternatives within 1% of the optimum, but supplier selections and allocation quantities can vary materially. Supplier and allocation recommendations should therefore be read as representatives of a near-optimal portfolio set rather than as unique prescriptions; this is precisely why the diagnostic is reported alongside the cost and emissions outcomes.

The nonlinear formulation is transformed into an explicit mixed-integer linear formulation by linearizing the buffer-lead-time and supplier-buffer interactions, enforcing supplier eligibility, and linking selection to positive allocated quantities. The focused validation demonstrates matching objectives within solver tolerance for all jointly optimal PLM/NLM test pairs, while the PLM proves optimality rapidly for every tested stress case up to 150 components. The study therefore supports an empirically validated linearized formulation and deterministic cross-scenario behavior, not universal equivalence or stochastic resilience.

Future work should test the resulting system against demand variabil-

ity, supplier disruptions and execution dynamics to evaluate whether its DDMRP-based flow-protection intent translates into operational resilience. Additional extensions include richer supplier quality and reliability attributes, jointly optimizing the emissions price and cap stringency, and validating the framework with industrial data.

Appendix A. Detailed numerical results

This appendix gathers the complete results that complement the representative samples discussed in Section 5 and the results subsections.

Appendix A.1. Full hybrid strategy results

Table A.9 reports the complete full-factorial combined carbon-price-and-cap results for every representative instance, with *EmisTax* in constant 2023 EUR/tCO₂ and the cap expressed as a percentage of the lexicographic baseline emissions or as an explicit no-cap level. All 160 tested combinations are feasible and solved to optimality; the achieved emissions vary with the cap level and are invariant to the carbon price, as discussed in Section 5.

Table A.9: Full results of the hybrid strategy across all tested instances, for every combination of carbon price *EmisTax* (currency/tCO₂) and emission cap.

Instance	<i>EmisTax</i>	Cap level	Emissions (t CO ₂)	Cost	Buffers	Status
bom_5	0.00	2.05	1.50	60 260	5	OPTIMAL
bom_5	0.00	2.20	1.50	60 260	5	OPTIMAL
bom_5	0.00	2.35	2.25	51 555	4	OPTIMAL
bom_5	0.00	2.49	2.25	51 555	4	OPTIMAL
bom_5	0.00	2.64	2.25	51 555	4	OPTIMAL
bom_5	0.00	2.79	2.65	48 815	3	OPTIMAL
bom_5	0.00	2.93	2.93	48 640	3	OPTIMAL
bom_5	0.00	No cap	2.93	48 640	3	OPTIMAL
bom_5	15.00	2.05	1.50	60 283	5	OPTIMAL
bom_5	15.00	2.20	1.50	60 283	5	OPTIMAL
bom_5	15.00	2.35	2.25	51 589	4	OPTIMAL
bom_5	15.00	2.49	2.25	51 589	4	OPTIMAL
bom_5	15.00	2.64	2.25	51 589	4	OPTIMAL
bom_5	15.00	2.79	2.65	48 855	3	OPTIMAL
bom_5	15.00	2.93	2.93	48 684	3	OPTIMAL
bom_5	15.00	No cap	2.93	48 684	3	OPTIMAL
bom_5	50.00	2.05	1.50	60 335	5	OPTIMAL
bom_5	50.00	2.20	1.50	60 335	5	OPTIMAL
bom_5	50.00	2.35	2.25	51 668	4	OPTIMAL
bom_5	50.00	2.49	2.25	51 668	4	OPTIMAL
bom_5	50.00	2.64	2.25	51 668	4	OPTIMAL
bom_5	50.00	2.79	2.65	48 947	3	OPTIMAL

Continued on next page

Instance	<i>EmisTax</i>	Cap level	Emissions (t CO ₂)	Cost	Buffers	Status
bom_5	50.00	2.93	2.93	48 787	3	OPTIMAL
bom_5	50.00	No cap	2.93	48 787	3	OPTIMAL
bom_5	75.00	2.05	1.50	60 373	5	OPTIMAL
bom_5	75.00	2.20	1.50	60 373	5	OPTIMAL
bom_5	75.00	2.35	2.25	51 724	4	OPTIMAL
bom_5	75.00	2.49	2.25	51 724	4	OPTIMAL
bom_5	75.00	2.64	2.25	51 724	4	OPTIMAL
bom_5	75.00	2.79	2.65	49 013	3	OPTIMAL
bom_5	75.00	2.93	2.93	48 860	3	OPTIMAL
bom_5	75.00	No cap	2.93	48 860	3	OPTIMAL
bom_5	100.00	2.05	1.50	60 410	5	OPTIMAL
bom_5	100.00	2.20	1.50	60 410	5	OPTIMAL
bom_5	100.00	2.35	2.25	51 780	4	OPTIMAL
bom_5	100.00	2.49	2.25	51 780	4	OPTIMAL
bom_5	100.00	2.64	2.25	51 780	4	OPTIMAL
bom_5	100.00	2.79	2.65	49 080	3	OPTIMAL
bom_5	100.00	2.93	2.93	48 933	3	OPTIMAL
bom_5	100.00	No cap	2.93	48 933	3	OPTIMAL
bom_13	0.00	20.6	20.0	170 516	10	OPTIMAL
bom_13	0.00	22.1	22.1	151 636	10	OPTIMAL
bom_13	0.00	23.6	23.6	127 888	9	OPTIMAL
bom_13	0.00	25.1	25.0	127 100	9	OPTIMAL
bom_13	0.00	26.5	26.1	119 556	7	OPTIMAL
bom_13	0.00	28.0	27.9	118 411	7	OPTIMAL
bom_13	0.00	29.5	29.5	116 448	7	OPTIMAL
bom_13	0.00	No cap	29.5	116 448	7	OPTIMAL
bom_13	15.00	20.6	20.0	170 815	10	OPTIMAL
bom_13	15.00	22.1	22.1	151 967	10	OPTIMAL
bom_13	15.00	23.6	23.6	128 241	9	OPTIMAL
bom_13	15.00	25.1	25.0	127 475	9	OPTIMAL
bom_13	15.00	26.5	26.1	119 948	7	OPTIMAL
bom_13	15.00	28.0	27.9	118 829	7	OPTIMAL
bom_13	15.00	29.5	29.5	116 890	7	OPTIMAL
bom_13	15.00	No cap	29.5	116 890	7	OPTIMAL
bom_13	50.00	20.6	20.0	171 514	10	OPTIMAL
bom_13	50.00	22.1	22.1	152 740	10	OPTIMAL
bom_13	50.00	23.6	23.6	129 066	9	OPTIMAL
bom_13	50.00	25.1	25.0	128 350	9	OPTIMAL
bom_13	50.00	26.5	26.1	120 863	7	OPTIMAL
bom_13	50.00	28.0	27.9	119 805	7	OPTIMAL
bom_13	50.00	29.5	29.5	117 923	7	OPTIMAL
bom_13	50.00	No cap	29.5	117 923	7	OPTIMAL
bom_13	75.00	20.6	20.0	172 013	10	OPTIMAL
bom_13	75.00	22.1	22.1	153 292	10	OPTIMAL
bom_13	75.00	23.6	23.6	129 655	9	OPTIMAL
bom_13	75.00	25.1	25.0	128 975	9	OPTIMAL
bom_13	75.00	26.5	26.1	121 517	7	OPTIMAL
bom_13	75.00	28.0	27.9	120 501	7	OPTIMAL
bom_13	75.00	29.5	29.5	118 660	7	OPTIMAL
bom_13	75.00	No cap	29.5	118 660	7	OPTIMAL
bom_13	100.00	20.6	20.0	172 512	10	OPTIMAL
bom_13	100.00	22.1	22.1	153 844	10	OPTIMAL
bom_13	100.00	23.6	23.6	130 244	9	OPTIMAL
bom_13	100.00	25.1	25.0	129 600	9	OPTIMAL

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Instance	<i>EmisTax</i>	Cap level	Emissions (t CO ₂)	Cost	Buffers	Status
bom_13	100.00	26.5	26.1	122 171	7	OPTIMAL
bom_13	100.00	28.0	27.9	121 198	7	OPTIMAL
bom_13	100.00	29.5	29.5	119 398	7	OPTIMAL
bom_13	100.00	No cap	29.5	119 398	7	OPTIMAL
bom_26	0.00	29.1	29.0	1 008 240	1	OPTIMAL
bom_26	0.00	31.2	31.1	957 140	1	OPTIMAL
bom_26	0.00	33.3	33.0	931 800	1	OPTIMAL
bom_26	0.00	35.3	35.3	909 180	1	OPTIMAL
bom_26	0.00	37.4	37.2	888 160	1	OPTIMAL
bom_26	0.00	39.5	39.2	874 160	1	OPTIMAL
bom_26	0.00	41.6	41.6	867 120	1	OPTIMAL
bom_26	0.00	No cap	41.6	867 120	1	OPTIMAL
bom_26	15.00	29.1	29.0	1 008 675	1	OPTIMAL
bom_26	15.00	31.2	31.1	957 607	1	OPTIMAL
bom_26	15.00	33.3	33.0	932 295	1	OPTIMAL
bom_26	15.00	35.3	35.3	909 709	1	OPTIMAL
bom_26	15.00	37.4	37.2	888 719	1	OPTIMAL
bom_26	15.00	39.5	39.2	874 748	1	OPTIMAL
bom_26	15.00	41.6	41.6	867 744	1	OPTIMAL
bom_26	15.00	No cap	41.6	867 744	1	OPTIMAL
bom_26	50.00	29.1	29.0	1 009 691	1	OPTIMAL
bom_26	50.00	31.2	31.1	958 697	1	OPTIMAL
bom_26	50.00	33.3	33.0	933 451	1	OPTIMAL
bom_26	50.00	35.3	35.3	910 944	1	OPTIMAL
bom_26	50.00	37.4	37.2	890 022	1	OPTIMAL
bom_26	50.00	39.5	39.2	876 120	1	OPTIMAL
bom_26	50.00	41.6	41.6	869 199	1	OPTIMAL
bom_26	50.00	No cap	41.6	869 199	1	OPTIMAL
bom_26	75.00	29.1	29.0	1 010 416	1	OPTIMAL
bom_26	75.00	31.2	31.1	959 476	1	OPTIMAL
bom_26	75.00	33.3	33.0	934 276	1	OPTIMAL
bom_26	75.00	35.3	35.3	911 827	1	OPTIMAL
bom_26	75.00	37.4	37.2	890 953	1	OPTIMAL
bom_26	75.00	39.5	39.2	877 100	1	OPTIMAL
bom_26	75.00	41.6	41.6	870 238	1	OPTIMAL
bom_26	75.00	No cap	41.6	870 238	1	OPTIMAL
bom_26	100.00	29.1	29.0	1 011 141	1	OPTIMAL
bom_26	100.00	31.2	31.1	960 254	1	OPTIMAL
bom_26	100.00	33.3	33.0	935 102	1	OPTIMAL
bom_26	100.00	35.3	35.3	912 709	1	OPTIMAL
bom_26	100.00	37.4	37.2	891 884	1	OPTIMAL
bom_26	100.00	39.5	39.2	878 080	1	OPTIMAL
bom_26	100.00	41.6	41.6	871 277	1	OPTIMAL
bom_26	100.00	No cap	41.6	871 277	1	OPTIMAL
bom_50	0.00	136.2	136.1	1 636 980	6	OPTIMAL
bom_50	0.00	145.9	145.9	1 601 800	6	OPTIMAL
bom_50	0.00	155.7	155.6	1 566 140	6	OPTIMAL
bom_50	0.00	165.4	165.3	1 497 760	5	OPTIMAL
bom_50	0.00	175.1	175.0	1 436 340	5	OPTIMAL
bom_50	0.00	184.8	184.8	1 402 860	5	OPTIMAL
bom_50	0.00	194.6	194.6	1 381 460	5	OPTIMAL
bom_50	0.00	No cap	194.6	1 381 460	5	OPTIMAL
bom_50	15.00	136.2	136.1	1 639 022	6	OPTIMAL
bom_50	15.00	145.9	145.9	1 603 988	6	OPTIMAL

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Instance	<i>EmisTax</i>	Cap level	Emissions (t CO ₂)	Cost	Buffers	Status
bom_50	15.00	155.7	155.6	1 568 474	6	OPTIMAL
bom_50	15.00	165.4	165.3	1 500 239	5	OPTIMAL
bom_50	15.00	175.1	175.0	1 438 965	5	OPTIMAL
bom_50	15.00	184.8	184.8	1 405 631	5	OPTIMAL
bom_50	15.00	194.6	194.6	1 384 379	5	OPTIMAL
bom_50	15.00	No cap	194.6	1 384 379	5	OPTIMAL
bom_50	50.00	136.2	136.1	1 643 787	6	OPTIMAL
bom_50	50.00	145.9	145.9	1 609 094	6	OPTIMAL
bom_50	50.00	155.7	155.6	1 573 918	6	OPTIMAL
bom_50	50.00	165.4	165.3	1 506 023	5	OPTIMAL
bom_50	50.00	175.1	175.0	1 445 089	5	OPTIMAL
bom_50	50.00	184.8	184.8	1 412 098	5	OPTIMAL
bom_50	50.00	194.6	194.6	1 391 189	5	OPTIMAL
bom_50	50.00	No cap	194.6	1 391 189	5	OPTIMAL
bom_50	75.00	136.2	136.1	1 647 191	6	OPTIMAL
bom_50	75.00	145.9	145.9	1 612 741	6	OPTIMAL
bom_50	75.00	155.7	155.6	1 577 808	6	OPTIMAL
bom_50	75.00	165.4	165.3	1 510 154	5	OPTIMAL
bom_50	75.00	175.1	175.0	1 449 464	5	OPTIMAL
bom_50	75.00	184.8	184.8	1 416 717	5	OPTIMAL
bom_50	75.00	194.6	194.6	1 396 053	5	OPTIMAL
bom_50	75.00	No cap	194.6	1 396 053	5	OPTIMAL
bom_50	100.00	136.2	136.1	1 650 594	6	OPTIMAL
bom_50	100.00	145.9	145.9	1 616 388	6	OPTIMAL
bom_50	100.00	155.7	155.6	1 581 697	6	OPTIMAL
bom_50	100.00	165.4	165.3	1 514 285	5	OPTIMAL
bom_50	100.00	175.1	175.0	1 453 838	5	OPTIMAL
bom_50	100.00	184.8	184.8	1 421 359	5	OPTIMAL
bom_50	100.00	194.6	194.6	1 400 918	5	OPTIMAL
bom_50	100.00	No cap	194.6	1 400 918	5	OPTIMAL

Appendix A.2. Multi-objective Pareto fronts

The cost–emissions and cost–DIO Pareto fronts plotted in Figures 5 and 6 are tabulated in Tables A.10 and A.11.

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Table A.10: Cost-emissions Pareto points obtained by the ε -constraint method.

Instance	Emissions (t CO ₂)	Cost
<i>bom_5</i>	1.50	66 492
	1.50	60 260
	2.25	51 555
	2.65	48 815
	2.93	48 640
	3.22	48 640
<i>bom_13</i>	9.86	531 080
	12.4	314 818
	14.8	292 400
	18.2	202 882
	21.0	154 424
	23.9	127 568
	26.6	119 556
	29.5	116 448
<i>bom_26</i>	7.04	3 104 390
	12.3	2 216 150
	17.6	1 511 520
	23.0	1 237 300
	28.3	1 025 140
	33.6	925 260
	38.6	876 960
	44.2	867 120
	46.9	867 120

Table A.11: Cost–DIO Pareto points obtained by the ε -constraint method.

Instance	DIO (days)	Cost
<i>bom_5</i>	23	60 260
	27	51 555
	29	48 815
	31	48 640
<i>bom_13</i>	74	496 798
	83	314 864
	93	154 744
	103	145 300
	113	127 843
	122	118 088
	132	116 668
	138	116 448
	134	116 448
<i>bom_26</i>	146	3 104 390
	193	1 785 510
	240	1 278 720
	287	1 055 840
	334	947 680
	382	886 760
	428	868 260
	476	867 120
	484	867 120

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